

DSAA3073: Theories in Data Science

Week 2 Validation Sheet · From Shape to Feasible Learning

Coverage: Week 2

Duration: 30 minutes · Total: 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Which set is convex?

- (a) $\{(-1, 0), (1, 0)\}$
- (b) $\{x \in \mathbb{R}^2 : \|x\|_2 = 1\}$
- (c) $\{x \in \mathbb{R}^2 : x_1 + x_2 = 1\}$
- (d) $\{x \in \mathbb{R}^2 : 1 \leq \|x\|_2 \leq 2\}$

Solution. (c). Take arbitrary x, y satisfying $x_1 + x_2 = 1$ and $y_1 + y_2 = 1$, and let $\lambda \in [0, 1]$. The coordinate sum of their convex combination is

$$\lambda(x_1 + x_2) + (1 - \lambda)(y_1 + y_2) = \lambda + (1 - \lambda) = 1.$$

Hence the whole segment remains in the affine line. The other choices all admit a midpoint that leaves the set: in each case, suitable opposite points have midpoint $(0, 0)$.

Marking. 3 marks for (c); 2 for a valid segment argument or for identifying why an apparent curved/two-point alternative fails.

A2 [5 marks]. A differentiable convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies $\nabla f(w^*) = 0$. Which conclusion is guaranteed?

- (a) w^* is a global minimizer, but it need not be unique.
- (b) w^* is the unique global minimizer.
- (c) f must be constant.
- (d) The gradient method converges to w^* for every positive step size.

Solution. (a). The first-order lower bound for a differentiable convex function gives, for every y ,

$$f(y) \geq f(w^*) + \nabla f(w^*)^T (y - w^*) = f(w^*).$$

Thus w^* is globally minimizing. Convexity does not by itself make the minimizer unique, and the stationary-point certificate does not supply an unrestricted step-size convergence theorem.

Marking. 3 marks for (a); 2 for using the zero gradient in the global tangent lower bound and not claiming uniqueness.

A3 [5 marks]. What is justified immediately after computing $w_{t+1} = w_t - \eta \nabla f(w_t)$ for a declared $\eta > 0$?

- (a) The update has been executed.
- (b) The objective value has decreased.
- (c) The iterates converge to a unique minimizer.
- (d) The algorithm is polynomial time.

Solution. (a). The displayed rule is a state transition: once the current gradient and step size are known, it defines the next iterate. Decrease and convergence need additional objective and step-size conditions, while a polynomial-time claim also needs an encoded input and a runtime analysis.

Marking. 3 marks for (a); 2 for distinguishing execution of an update from a decrease, convergence, or runtime guarantee.

A4 [5 marks]. A program evaluates all 2^d binary candidates. Which conclusion follows from this fact alone?

- (a) The stated program is not polynomial time in variable d .
- (b) No polynomial-time algorithm exists for the problem.
- (c) The problem has a computational–statistical gap.
- (d) The sample size must also be exponential in d .

Solution. (a). Exhaustive enumeration takes exponentially many evaluations when d is a variable input parameter. This diagnoses the stated implementation only. It neither rules out a different structured algorithm nor supplies a sample-complexity statement or a same-task computational obstruction.

Marking. 3 marks for (a); 2 for limiting the conclusion to the declared implementation and scaling parameter.

Part B: Definition in use

B1 [20 marks]. Define

$$C = \{x \in \mathbb{R}^2 : |x_1| + 2|x_2| \leq 4 \text{ and } x_1 - x_2 = 1\}.$$

Decide whether C is convex. Give a proof from the definition using arbitrary points of C and an arbitrary interpolation parameter; a diagram or a few numerical checks is not sufficient.

Solution. The set is convex. Let arbitrary $u, v \in C$ and arbitrary $\lambda \in [0, 1]$ be given, and set $z = \lambda u + (1 - \lambda)v$. We must verify both defining constraints for this same point z .

First, apply the scalar triangle inequality coordinate by coordinate. Since λ and $1 - \lambda$ are nonnegative,

$$\begin{aligned} |z_1| + 2|z_2| &= |\lambda u_1 + (1 - \lambda)v_1| + 2|\lambda u_2 + (1 - \lambda)v_2| \\ &\leq \lambda(|u_1| + 2|u_2|) + (1 - \lambda)(|v_1| + 2|v_2|) \leq 4. \end{aligned}$$

The last inequality uses $u, v \in C$.

The affine equality is preserved exactly:

$$z_1 - z_2 = \lambda(u_1 - u_2) + (1 - \lambda)(v_1 - v_2) = \lambda + (1 - \lambda) = 1.$$

Thus z satisfies both constraints and belongs to C . Because the points and interpolation parameter were arbitrary, every segment between points of C remains in C ; hence C is convex.

Marking. 4 marks for arbitrary $u, v \in C$, $\lambda \in [0, 1]$, and the correct interpolated point; 8 for the weighted absolute-value/triangle-inequality argument; 5 for preservation of the affine equality; 3 for checking both constraints and concluding from the universal definition. Preserve method credit after a minor algebraic slip if both constraints are audited for an arbitrary interpolation.

Part C: Basic skills

C1 [7 marks]. Let $f(t) = t^2 - 2t$. For $x = 0$, $y = 3$, and $\lambda = \frac{1}{3}$, compute both sides of the convex-function chord inequality. Does this one chord satisfy the inequality, and what does that single calculation not prove?

Solution. The interpolated input is

$$\lambda x + (1 - \lambda)y = \left(\frac{1}{3}\right)0 + \left(\frac{2}{3}\right)3 = 2.$$

Therefore the left side is $f(2) = 2^2 - 2(2) = 0$. The endpoint values are $f(0) = 0$ and $f(3) = 9 - 6 = 3$, so the chord height is

$$\lambda f(0) + (1 - \lambda)f(3) = \left(\frac{1}{3}\right)0 + \left(\frac{2}{3}\right)3 = 2.$$

Thus this chord satisfies $0 \leq 2$. The calculation checks one admissible triple (x, y, λ) ; by itself it does not prove convexity, whose definition quantifies over every admissible triple.

Marking. 2 marks for the interpolated input; 2 for the function value there; 2 for the chord height and correct comparison; 1 for the universal-quantifier limitation.

C2 [8 marks]. For integer $N \geq 2$, a routine has runtime

$$T(N) = N^2 + N2^{\lceil \log_2 N \rceil}.$$

Give a polynomial big- O bound in the encoded input length N and justify it using an explicit bound on $2^{\lceil \log_2 N \rceil}$.

Solution. By the definition of the ceiling,

$$\log_2 N \leq \lceil \log_2 N \rceil < \log_2 N + 1.$$

Exponentiating base 2 gives

$$N \leq 2^{\lceil \log_2 N \rceil} < 2N.$$

Hence

$$T(N) < N^2 + N(2N) = 3N^2.$$

Therefore $T(N) = O(N^2)$, so the routine is polynomial time in the declared encoded input length N .

Marking. 3 marks for a valid ceiling/logarithm bound; 3 for substituting it into the runtime; 2 for the conclusion $O(N^2)$ and the polynomial-time classification.

Part D: Construction and counterexample

D1 [17 marks]. Construct a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ for which $f'(0) = 0$ but 0 is not a global minimizer. Verify both claims, then give a specific violated chord inequality showing that your example is not convex.

Solution. One simple construction is $f(x) = x^3$. Its derivative is $f'(x) = 3x^2$, so $f'(0) = 0$. However,

$$f(-1) = -1 < 0 = f(0),$$

so 0 is not a global minimizer.

It remains to provide the requested convexity witness. Choose the endpoints $x = -2$ and $y = 0$ with $\lambda = \frac{1}{2}$. Their midpoint is -1 . The convex-function inequality would require

$$f(-1) \leq \frac{1}{2}f(-2) + \frac{1}{2}f(0).$$

For this function the two sides are

$$f(-1) = -1, \quad \frac{1}{2}f(-2) + \frac{1}{2}f(0) = \frac{1}{2}(-8) + \frac{1}{2}(0) = -4.$$

The required inequality is false because $-1 > -4$. Thus the same example has a stationary point that is not globally minimizing and is demonstrably nonconvex. This also shows why the convexity hypothesis is essential in the Week 2 stationary-point certificate.

Marking. 5 marks for a valid differentiable construction; 3 for verifying $f'(0) = 0$; 3 for exhibiting a point with value below $f(0)$; 5 for a complete numerical chord violation; 1 for connecting the failure to the missing convexity hypothesis. Award equivalent constructions in full when all requested properties are verified.

D2 [18 marks]. A student claims that the pointwise minimum of two convex functions must be convex. Construct two convex functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ for which $h(x) = \min(f(x), g(x))$ is not convex. Verify that both inputs are convex and give a complete chord witness against h .

Solution. Choose

$$f(x) = (x - 1)^2, \quad g(x) = (x + 1)^2.$$

Both are convex. For example, for arbitrary $a, b \in \mathbb{R}$ and $\lambda \in [0, 1]$, the chord height minus the value at the interpolated input for $(x - c)^2$ equals

$$\lambda(1 - \lambda)(a - b)^2 \geq 0;$$

taking $c = 1$ proves convexity of f , and taking $c = -1$ proves convexity of g .

Now consider their pointwise minimum. At the two endpoints,

$$h(-1) = \min(4, 0) = 0, \quad h(1) = \min(0, 4) = 0.$$

At their midpoint,

$$h(0) = \min(1, 1) = 1.$$

Convexity at $\lambda = \frac{1}{2}$ would require

$$h(0) \leq \frac{1}{2}h(-1) + \frac{1}{2}h(1) = 0,$$

but $1 > 0$. Hence h is not convex, disproving the universal claim.

Marking. 5 marks for a workable pair of functions; 5 for a general proof that both inputs are convex; 6 for two endpoints, an interpolation parameter, and a correct violated chord inequality for the minimum; 2 for the explicit counterexample conclusion. Preserve credit for any different valid pair with complete verification.

Part E: A two-sided convexity audit

E1 [25 marks]. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Suppose that both f and the function $r(x) = -f(x)$ are convex on $\mathbb{R}$. Characterize exactly all functions f with this property and prove both directions of your characterization. A list of examples without a general proof is not sufficient.

Solution. The functions with the stated property are exactly the affine functions

$$f(x) = ax + b$$

for constants $a, b \in \mathbb{R}$.

We first prove necessity. Since f is differentiable and convex on the open convex domain $\mathbb{R}$, its first-order condition gives, for arbitrary $x, y \in \mathbb{R}$,

$$f(y) \geq f(x) + f'(x)(y - x). \quad (1)$$

The same condition must also be applied to $r = -f$. Its derivative is $r'(x) = -f'(x)$, so convexity of r gives

$$r(y) \geq r(x) + r'(x)(y - x),$$

or

$$-f(y) \geq -f(x) - f'(x)(y - x).$$

Multiplying by -1 reverses the inequality:

$$f(y) \leq f(x) + f'(x)(y - x). \quad (2)$$

Inequalities (1) and (2) hold for the same arbitrary pair x, y and have identical right sides. They therefore force equality:

$$f(y) = f(x) + f'(x)(y - x) \quad \text{for all } x, y \in \mathbb{R}. \quad (3)$$

Fixing $x = 0$ in (3) yields

$$f(y) = f(0) + f'(0)y.$$

Thus f is affine, with $a = f'(0)$ and $b = f(0)$. The quantifier in (3) is what rules out a function that is locally straight at one point but curved elsewhere.

For sufficiency, let $f(x) = ax + b$. For arbitrary x, y and $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) = \lambda f(x) + (1 - \lambda)f(y).$$

Hence the convexity inequality holds with equality, so f is convex. Moreover $-f(x) = -ax - b$ is also affine and satisfies the same equality; therefore $-f$ is convex. Both directions are proved, including constant functions as the case $a = 0$.

Marking. 3 marks for the exact affine characterization; 5 for selecting and correctly applying the global first-order lower bound to f ; 6 for applying it independently to $-f$, including its derivative and the reversed inequality; 4 for combining the two bounds into equality for arbitrary x, y ; 3 for fixing one base point and deriving the global affine form; 4 for the converse via chord equality for both f and $-f$, including the constant case. Recall of one inequality alone earns at most 5 marks; preserve downstream credit after an isolated sign slip when the two-sided strategy is otherwise coherent.

Part F: Encoding changes the runtime verdict

F1 [15 marks]. The same table-search routine is run under two input formats.

- In format E, the file explicitly lists one one-bit score for each of the 2^d binary strings, in a fixed order. Thus its encoded length is $N_E = \Theta(2^d)$, and the routine scans every listed score once.
- In format S, the file is a circuit description of length $N_S = \Theta(d^3)$. It computes the score of any requested string in time polynomial in d , and the routine requests all 2^d strings.

Classify the stated routine as polynomial or non-polynomial in the encoded input length under each format. Justify both classifications and explain why the phrase “it loops 2^d times” is not by itself an encoding-independent runtime verdict.

Solution. In format E, the input itself contains 2^d score bits. Scanning every listed score once takes $\Theta(2^d)$ operations, while $N_E = \Theta(2^d)$. Therefore the scan is $\Theta(N_E)$, which is linear and hence polynomial in the actual encoded input length. The expression is exponential in the auxiliary parameter d , but the file is already that large.

In format S, the circuit is succinct: $N_S = \Theta(d^3)$. Evaluating all strings takes

$$2^d \text{ poly}(d)$$

time. This is not polynomial in N_S . Indeed, any polynomial N_S^k is $\Theta(d^{3k})$, while 2^d eventually exceeds every fixed power of d ; multiplying by another polynomial in d does not change that conclusion. Thus the stated enumeration routine is superpolynomial in the succinct encoded length.

The two verdicts differ because polynomial time is defined relative to the number of encoded symbols or bits. A loop count written in terms of d must be compared with how d is represented and with how much information the input explicitly contains. Hence “ 2^d iterations” describes the loop, but does not by itself decide whether the loop is polynomial in the encoded input length.

Marking. 5 marks for relating the explicit scan to N_E and classifying it as linear; 6 for converting $N_S = \Theta(d^3)$ into a polynomial comparison and proving the succinct enumeration superpolynomial; 4 for the encoding-dependent explanation. Follow through consistently after a minor asymptotic-notation slip when the two representations are compared correctly.